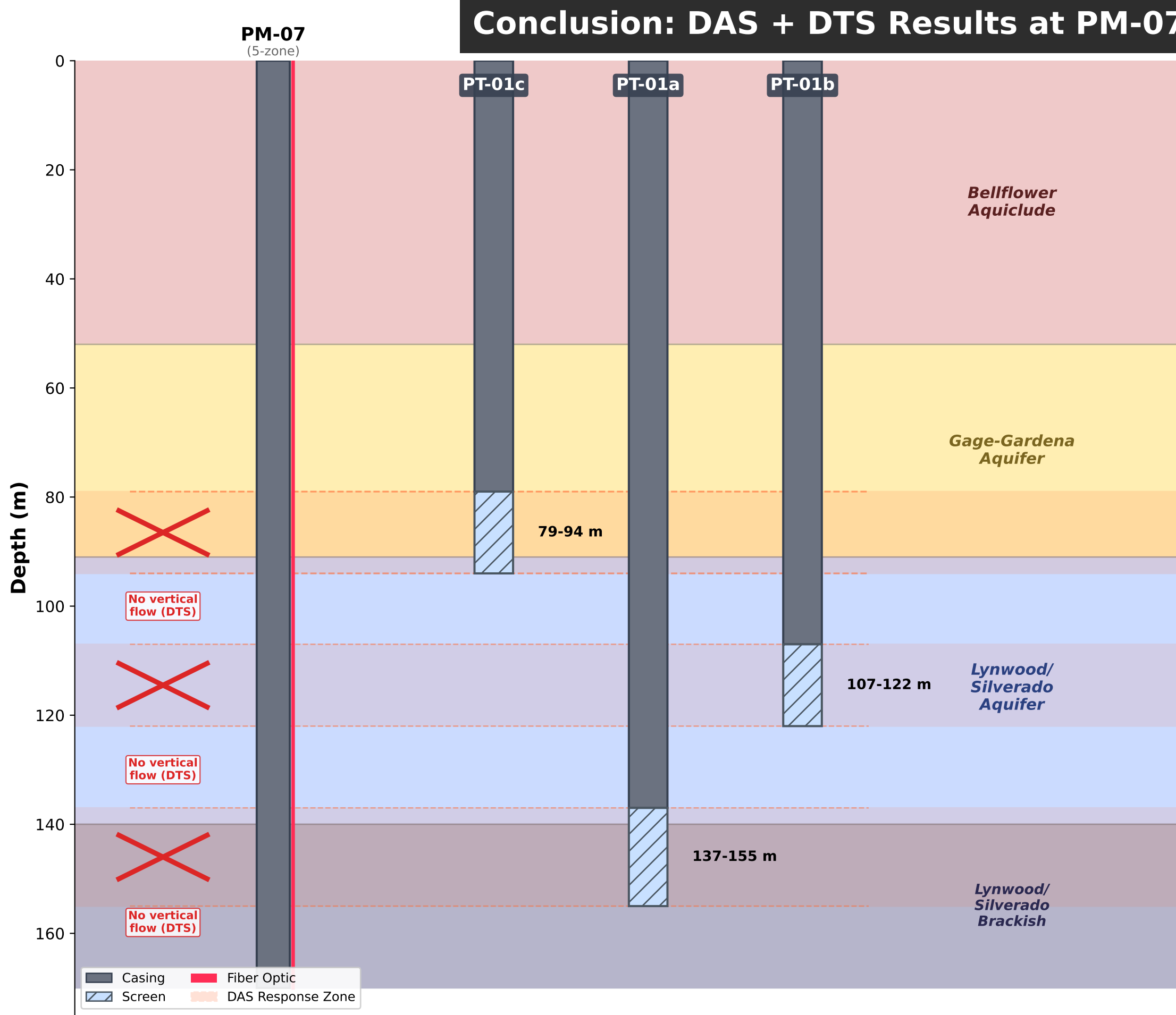


Conclusion: DAS + DTS Results at PM-07 Across Three Pump Tests



Storage Parameter Results

PT-01c (Gage-Gardena, 79-94 m)

$$S_s = 1.05 \times 10^{-7} \text{ 1/m}$$

$$R^2 = 0.941$$

Slope: 3.17×10^{-8} | Zone 5 piezometer

PT-01b (Lynwood-Silverado, 107-122 m)

$$S_s = 7.10 \times 10^{-8} \text{ 1/m}$$

$$R^2 = 0.976$$

Slope: 2.16×10^{-8} | Zone 4 piezometer

PT-01a (Lynwood-Silverado, 137-155 m)

$$S_s = 7.05 \times 10^{-8} \text{ 1/m}$$

$$R^2 = 0.971$$

Slope: 2.15×10^{-8} | Zone 2 piezometer

Key Findings

✓ Vertical stratification confirmed

DAS strain confined to pumped zones | DTS: no vertical flow

✓ Consistent storage across 76 m vertical span

S_s range: 7.05×10^{-8} to 1.05×10^{-7} 1/m (1.5× variation)

✓ Strong linear correlations ($R^2 = 0.94\text{--}0.98$)

Poroelastic response validated against independent piezometers

⚡ 2,600+ DAS points vs. 5 piezometer zones

Continuous strain profile at 0.25 m spacing across 665 ft

⚠ AQTESOLV discrepancy: 243–391× higher S_s

DAS = elastic skeletal storage | AQTESOLV = integrated hydraulic response

One fiber optic cable turned a 5-point pressure profile into a continuous picture of aquifer response